The integer anchor matters at small noise

Research note, 28 October 2026

The central asymmetric comparison cannot be extended to all thresholds by always taking the integer part of the raw threshold. The following example uses an actual iid three-point law, an exact atom, and rational bounds on the normal integral. It also explains why the retained shifted-anchor theorem is unaffected. A second result replaces a square-root noise term in the binomial curvature estimate by actual first absolute moments.

1 A counterexample to a fixed floor-anchor extension

For a centered Bernoulli-coordinate law $Y = B + U$, write

$$v = p(1-p), \quad s = \mathbb{E}U^2, \quad V = v + s, \quad \rho = \mathbb{E}|Y-p|^3/V^{3/2}, \quad \rho_B = (1-2v)/\sqrt{v}.$$

At raw threshold t, define the comparison with integer anchor j by

$$\Delta_C(t,j) = \sqrt{n}\left\{\mathbb{P}(Y_1+\cdots+Y_n \le t) - F_B(j) - \Phi\left(\frac{t-np}{\sqrt{nV}}\right) + \Phi\left(\frac{j-np}{\sqrt{nv}}\right)\right\} + C(\rho_B - \rho),$$

where the Y_i are independent copies and F_B is the binomial CDF.

Theorem 1. *Take $n = 42$, $p = 1/2$, $A = 3/5$, $\varepsilon = 1/350000$, $U_0 = 0$, and the centered conditional law $U_1 \in \{-\varepsilon, A\}$. Put*

$$h = 21\varepsilon = 3/50000, \qquad t = 21 - h.$$

Then $\lambda = ns = 9/250000$, the inward displacement is $\varepsilon < .072$, and $|H| = |t-21|/\sqrt{V} < 1.84$. For every $0 \le C \le 1$,

$$\Delta_C(t, \lfloor t \rfloor) > 1/250.$$

In contrast, $\Delta_C(t, 21) < 0$ for every $C \ge 0$.

Proof. The probability of the rare positive conditional jump is $\theta = \varepsilon/(A+\varepsilon) = 1/210001$. If the binomial count is at most 21 and none of its conditional jumps is rare, the sum is at most $21(1-\varepsilon) = t$. A union bound therefore gives

$$\mathbb{P}(Y_1+\cdots+Y_n \le t) \ge F_B(21) - 21\theta.$$

Here $\lfloor t \rfloor = 20$ and $\pi_{21} = \binom{42}{21}/2^{42}$.

Since $t < 21$, the normal interval in the comparison is at most $\Phi(0) - \Phi(-a)$, where $a = \sqrt{2/21}$. Using $e^{-x^2/2} \le 1 - x^2/2 + x^4/8$ and $\phi(0) < 2/5$ yields

$$\sqrt{42}\{\Phi(0) - \Phi(-a)\} \le 2\phi(0)\left(1 - \frac{1}{63} + \frac{1}{4410}\right) < \frac{4}{5}\frac{1447}{1470}.$$

The standard constant bound $\phi(0) < 2/5$ follows from $\pi > 25/8$. For example Machin's identity and alternating arctangent bounds give the strict rational lower bound $\pi > 16(1/5 - 1/375) - 4/239 > 25/8$.

Put $w = A\varepsilon$. The raw third absolute moment is exactly

$$M_3 = \frac{1}{8} + \frac{3}{4}w + \frac{1}{2}w(A - \varepsilon).$$

Since $V \geq 1/4$, $\rho \leq 8M_3$, while $\rho_B = 1$. Thus for $0 \leq C \leq 1$ the penalty is at least $-(8M_3 - 1)$. Finally $162/25 < \sqrt{42} < 649/100$. Consequently

$$\begin{aligned}\Delta_C(t, 20) &> \frac{162}{25}\pi_{21} - \frac{649}{100}\,21\theta - \frac{4}{5}\frac{1447}{1470} - (8M_3 - 1) \\ &= \frac{185046727061827375519549}{376665544851456000000000000} > \frac{1}{250}.\end{aligned}$$

Every number on the last line is rational. No computed normal CDF is used. The variance and raw threshold bounds follow by squaring positive quantities.

For the other anchor, a count of at least 22 cannot contribute even if all its jumps are short. At count 21, at least one rare jump moves the sum strictly above t. Thus

$$\mathbb{P}(Y_1 + \cdots + Y_n \leq t) \leq F_B(21) - \pi_{21}\{1 - (1 + 1/210000)^{-21}\}.$$

Bernoulli's inequality bounds the bracket below by $1/10001$. The adverse normal change from the central anchor is less than $\phi(0)h/\sqrt{V} < (4/5)h$. Jensen gives $\rho \geq 1$, so for every $C \geq 0$

$$\Delta_C(t, 21) < -\frac{162}{25}\frac{\pi_{21}}{10001} + \frac{4}{5}h < 0.$$

The last strict inequality is another rational check. The event at t is closed throughout: the common short-jump configuration lies exactly at it. □

The sharp Bernoulli coefficient $C_E = (3 + \sqrt{10})/(6\sqrt{2\pi})$ lies in $(0, 1)$, so the counterexample includes that coefficient. It is a counterexample to a proposed choice of comparison anchor, not to the Berry–Esseen conjecture.

2 Audit of the retained anchor rules

The retained v12 union theorem explicitly supplies an integer anchor. Its right-hand proof chooses phase $u = t - j \in [-.272, .728]$; equivalently one may choose $j = \lfloor t + .272 \rfloor$. The v11 theorem states the same phase interval. In Theorem 1 this anchor is 21 and its phase is $-3/50000$, exactly the anchor proved favorable above.

The v14 larger-noise theorem explicitly takes $j = \lfloor t \rfloor$, but assumes $\lambda \geq .075$. The present noise is $9/250000 < .075$, so it does not meet that theorem's hypotheses. The v14 join with the smaller-noise result remains valid: the external strict Bernoulli theorem holds at every integer anchor. The join needs existence of a suitable anchor in each component, not agreement of the two anchors. No retained theorem is refuted; any prose or implementation identifying these anchors uniformly would need correction.

For a fixed actual law and threshold, changing anchors gives exactly

$$\Delta_C(t, 20) - \Delta_C(t, 21) = \sqrt{42}\{\pi_{21} - \Phi(0) + \Phi(-\sqrt{2/21})\}.$$

This positive, noise-independent quantity explains why a small-defect proof cannot absorb a wrong anchor in an error merely proportional to the noise.

3 A curvature bound using first absolute moments

Let $v_j = \mathbb{E}U_j^2$, $e_j = \mathbb{E}(U_j)_+ = \mathbb{E}(U_j)_-$, and $\ell_1 = (1-p)e_0 + pe_1$. For any real t, set

$$B_j = \mathbb{E}(t-j-W_j)_+ - (t-j)_+, \quad \mathcal{C}(t) = \sum_{j=1}^{n-1}(\pi_{j+1}-\pi_j)B_j, \quad c = (np-(1-p)-t)_+.$$

Lemma 2. *For centered independent conditional summands of finite second moments,*

$$\sqrt{n}\,\mathcal{C}(t) \le \frac{\lambda}{4v\sqrt{n}} + \frac{c}{v}\min\left\{\frac{\sqrt{\lambda}}{2\sqrt{n+1}}, \sqrt{n}\,\ell_1\right\}.$$

For either one-noisy conditional two-point family with outward size A and inward size ε,

$$\ell_1 = \frac{s}{A+\varepsilon}, \qquad \sqrt{n}\,\mathcal{C}(t) \le \frac{\lambda}{4v\sqrt{n}} + \frac{c\lambda}{v(A+\varepsilon)\sqrt{n}}.$$

The second remainder is linear in arbitrarily small positive noise.

Proof. The retained v13 proof gives $B_j \ge 0$ and $(t-j)_+B_j \le \mathrm{Var}(W_j)/4$. Also

$$B_j \le \mathbb{E}(W_j)_+ \le (n-j)e_0 + je_1.$$

For the first inequality use put–call parity separately for $t-j \ge 0$ and $t-j \le 0$; for the second use subadditivity of the positive part. Write $\widetilde{\pi}_r$ for the $\mathrm{Bin}(n+1,p)$ masses. The exact binomial difference formula is

$$\pi_{j+1} - \pi_j = \frac{np-(1-p)-j}{(n+1)v}\widetilde{\pi}_{j+1}.$$

Discard negative coefficients, and bound the positive numerator by $(t-j)_+ + c$. The variance part sums to at most $(n+1)s/4$. With $r = j+1$, the other part is bounded by the expectation of $(n+1-r)e_0 + re_1$ over the full binomial range, hence by $(n+1)\ell_1$. Therefore $\mathcal{C} \le s/(4v) + c\ell_1/v$. Combine this with the independently retained square-root curvature bound. For a centered two-point variable on $\{-A, \varepsilon\}$, or its reflection, its variance is $A\varepsilon$ and each one-sided first moment is $A\varepsilon/(A+\varepsilon)$. This proves the final specialization. □

The companion exact checks verify the counterexample, the favorable shifted anchor, and the new curvature inequality on actual laws, including two noisy conditional laws, tiny defects and exact atom collisions. A bounded floating exploration of moderate-noise phases found no failure among its tested laws; that observation is discovery evidence only and proves no uniform phase domain.